For example, if a user enters 100 as input; initially, 100 is stored in the variable **n**. In the next statement, the value of **n** i.e., 100 is assigned to **ch**. Since, **ch** is a char type; it shows the corresponding ASCII character as output. (Equivalent ASCII Character for 100 is d).

(3) float data type:

If a variable is declared as float, all values will be stored as floating point values.

There are two advantages of using float data types.

(1) They can represent values between the integers.

(2) They can represent a much greater range of values.

At the same time, floating point operations takes more time to execute compared to the integer type ie., floating point operations are slower than integer operations. This is a disadvantage of floating point operation.

For Example

```
float num=13.4;
```

In the above example, num variable is declared as float type .

(4) double data type:

This is for double precision floating point numbers. (precision means significant numbers after decimal point). The double is also used for handling floating point numbers. But, this type occupies double the space than float type. This means, more fractions can be accommodated in double than in float type. The double is larger and slower than type float. double is used in a similar way as that of float data type.

(5) void data type:

The literal meaning for void is 'empty space'. Here, in C++, the void data type specifies an empty set of values. It is used as a return type for functions that do not return any value.

Evaluate Yourself

1. What do you mean by fundemantal data types?
2. The data type char is used to represent characters. then why is it often termed as an integer type?
3. What is the advantage of floating point numbers over integers?
4. The data type double is another floating point type. Why is it treated as a distinct data type?
5. What is the use of void data type?

9.12.2 Memory representation of Fundamental Data types:

One of the most important reason for declaring a variable as a particular data type is to allocate appropriate space in memory. As per the stored program concept, every data should be accommodated in the main memory before they are processed. So, C++ compiler allocates specific memory space for each and every data handled according to

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